PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Test Plan (STP)

Hospital Management System

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version
Navya Suresh (PES1UG24CS904)	02-10-2026	Initial draft for Mini-Project Part-1	1.0

1. Introduction

Purpose: This document defines the test plan for the Hospital Management System (HMS) v1.0. It states the objectives, scope, strategy, resources, schedule and responsibilities for testing, and it lists the test cases that verify the functional, non-functional and security requirements of the SRS.

Scope: Testing covers authentication and access control, patient management, appointment management, clinical records, billing, reports and administration, together with performance, availability, usability, reliability, portability, maintainability and security. Hospital network hardware, the operating system and third-party library internals are excluded.

References: HMS SRS v1.0 (SRS_Document.docx), HMS Software Architecture and Design Specification v1.0, IEEE Std 829-2008 (software test documentation), OWASP ASVS 4.0 and OWASP Testing Guide, DPDP Act 2023.

Definitions: HMS (Hospital Management System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), PHI (Protected Health Information), TLS (Transport Layer Security), RBAC (Role-Based Access Control).

2. Test Items

- Authentication and access control module (Auth Service)
- Patient management module (Patient Service)
- Appointment management module (Appointment Service)
- Clinical records module (Clinical Service)
- Billing module (Billing Service)
- Report module (Report Service)
- Administration and audit module (Audit Logger)
- REST API controller and Data Access Layer
- Desktop UI client (staff console and patient portal)
- Backup job

3. Features to be Tested

Features are mapped to SRS requirement IDs (full traceability in Section 13).

Feature group	Requirement IDs
Authentication and Access Control	HMS-F-001, HMS-F-002, HMS-F-003, HMS-F-004
Patient Management	HMS-F-010, HMS-F-011, HMS-F-012, HMS-F-013, HMS-F-014
Appointment Management	HMS-F-020, HMS-F-021, HMS-F-022, HMS-F-023, HMS-F-024
Clinical Records	HMS-F-030, HMS-F-031, HMS-F-032, HMS-F-033, HMS-F-034
Billing	HMS-F-040, HMS-F-041, HMS-F-042
Reports and Administration	HMS-F-050, HMS-F-060, HMS-F-061
Performance and capacity	HMS-NF-001, HMS-NF-002, HMS-NF-003
Availability and reliability	HMS-NF-004, HMS-NF-006
Usability, portability, maintainability	HMS-NF-005, HMS-NF-009, HMS-NF-007, HMS-NF-008
Security	HMS-SEC-001, HMS-SEC-002, HMS-SEC-003, HMS-SEC-004, HMS-SEC-005, HMS-SEC-006, HMS-SEC-007, HMS-SEC-008

4. Features Not to be Tested

- Internals of PostgreSQL, OpenSSL, Qt and the HTTP library (assumed tested by their vendors; only our use of them is tested).
- Hospital network infrastructure and server hardware.
- Features outside the SRS scope: inpatient beds, pharmacy stock, laboratory, insurance claims, payment gateways.

- Optional SMS / e-mail notification adapter (not part of release 1.0 acceptance).

5. Test Approach / Strategy

Levels

- **Unit tests** for each service and repository class using GoogleTest, with gcov/lcov for coverage (target 70% or more).
- **Integration tests** between API controller, services and PostgreSQL (test database), using Postman / Newman collections.
- **System tests** of end-to-end flows (register, book, consult, prescribe, bill, pay) through the desktop client.
- **Acceptance tests (UAT)** by the course instructor / a nominated hospital user against the SRS acceptance criteria.

Types

- Functional testing (equivalence classes and boundary values for fields, dates, amounts and time limits).
- Negative testing (invalid input, wrong role, expired session).
- Regression testing: the automated suite is re-run after every merged change.
- Performance and load testing (response time, 200 concurrent users) with JMeter.
- Usability testing with 5 trained users; compatibility testing on Windows 11 and Ubuntu 22.04.
- Reliability / recovery testing (backup and restore drill).

Entry criteria: stable build delivered and tagged, test database with dummy data loaded, test environment ready, SRS baselined. **Exit criteria:** 100% of planned test cases executed, 95% passed, 0 critical or high defects open, every requirement covered by at least one passed test case, acceptance criteria satisfied.

5.1 Security Validation

Each security requirement (HMS-SEC-001 to 008) is validated as follows. Test cases are listed in Appendix A.

Area	Validation activity	Tool	Test case
Authentication and passwords	Password policy enforced; credentials table inspected for salted hashes; lock-out after 5 failures	Manual, SQL client	TC-SEC-01, TC-AUTH-03
Transport security	TLS 1.2+ only, plain HTTP refused, weak ciphers absent	testssl.sh, Wireshark	TC-SEC-02
Session management	Idle timeout 10 min; token invalid after logout; no token reuse	Postman, Burp Suite	TC-SEC-03
Input validation and injection	SQL injection, script injection and over-long input on all fields; fuzzing of API JSON fields	OWASP ZAP, sqlmap, manual	TC-SEC-04
Access control	Role and object-level checks on every endpoint (horizontal and vertical privilege escalation)	Postman, Burp Suite	TC-RBAC-01, TC-SEC-07
Audit and logging	Audit entries exist for actions; logs contain no passwords or PHI	Manual log review	TC-SEC-05
Data protection at rest	Sensitive columns are ciphertext; backups encrypted	SQL client	TC-SEC-06, TC-REL-01
Penetration test	Authentication and authorisation flows tested by a team member who did not write the code	OWASP ZAP	TC-SEC-01 to 07

6. Test Environment

Hardware: one server (4 cores, 8 GB RAM), two client PCs (Windows 11 and Ubuntu 22.04), one printer. **Software:** HMS server and client v1.0, PostgreSQL 15 test database, OpenSSL 3. **Tools:** GoogleTest and gcov/lcov (unit and coverage), Postman / Newman (API), JMeter (performance), OWASP ZAP, Burp Suite Community and testssl.sh (security), Git and a spreadsheet or issue tracker (defects). **Test data:** 100,000 dummy patients, 20 doctors with schedules, dummy users for each role, sample bills. No real patient data is used.

7. Test Schedule

Milestone	Planned date
Test case design (this document)	05-Oct-2026

Milestone	Planned date
Plan and test cases submitted (Part-1)	07-Oct-2026
Environment setup and test data load	20-Oct-2026
Test execution start	26-Oct-2026
Test execution end	06-Nov-2026
UAT	09-Nov-2026 to 11-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases (Appendix A: manual and automated)
- Test scripts (GoogleTest, Newman, JMeter)
- Test data
- Test execution logs
- Defect reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Daksha S G (PES1UG24CS137)	Prepare plan, coordinate execution, report status
Test Engineer	Gnanasagar H (PES1UG24CS171)	Design and execute test cases, log defects
Developer	Dhruv Swatantramath (PES1UG24CS155)	Fix defects, support triage, write unit tests
Product Owner	Navya Suresh (PES1UG24CS904)	Approve test results and sign off readiness

Group 11. Roles within the test team can be rotated.

10. Risks and Mitigation

Risk	Mitigation
Late delivery of a stable build	Request weekly smoke builds; run the automated suite on every merge
Test environment downtime	Keep a second environment as a virtual machine image and database dump
Performance targets not met at 100,000 records	Run a first load test early; add database indexes; tune queries
Security defects found late	Run OWASP ZAP baseline scan weekly; review code for injection risks early
Small team, limited time	Prioritise High-priority requirements first; automate regression tests

11. Assumptions & Dependencies

- Test database with dummy patients, doctors and users is prepared before execution starts.
- The server and clients stay available during the schedule.
- Open-source tools (JMeter, ZAP, GoogleTest) are installable on the test machines.
- Requirements in SRS v1.0 are frozen during test execution.

12. Suspension & Resumption Criteria

Suspend testing if: the environment is unavailable for more than 4 hours; the build is so unstable that more than 30% of test cases are blocked; or a critical security defect (for example, authentication bypass) is found. **Resume testing when:** blocking defects are fixed and verified by a smoke test, and the environment is stable again.

13. Test Case Management & Traceability

The Requirements Traceability Matrix below maps every SRS requirement to its test cases. The reverse mapping (test case to requirement) is the "Requirement" column of Appendix A. Every requirement has at least one test case.

Requirement ID	Short description	Test case(s)
HMS-F-001	Authenticate every user with a unique username and password	TC-AUTH-01
HMS-F-002	Lock account for 15 minutes after 5 failed attempts	TC-AUTH-03
HMS-F-003	Enforce role-based access (role-permission matrix)	TC-RBAC-01
HMS-F-004	Generic message on invalid credentials	TC-AUTH-02
HMS-F-010	Register a new patient	TC-PAT-01
HMS-F-011	Reject invalid registration, name the offending field	TC-PAT-02
HMS-F-012	Warn on duplicate patient before saving	TC-PAT-03
HMS-F-013	Search patients by ID, name or phone	TC-PAT-04
HMS-F-014	Update demographics, retain previous values	TC-PAT-05
HMS-F-020	Book an appointment	TC-APT-01
HMS-F-021	Prevent double-booking, show next 3 slots	TC-APT-02
HMS-F-022	Reject past and more-than-60-day bookings	TC-APT-03
HMS-F-023	Reschedule or cancel up to 2 hours before start	TC-APT-04
HMS-F-024	Show doctor's daily appointment list	TC-APT-05
HMS-F-030	Record symptoms, diagnosis and notes for a visit	TC-CLN-01
HMS-F-031	Add prescription items to a visit	TC-CLN-02
HMS-F-032	Allergy warning with override reason	TC-CLN-03
HMS-F-033	Display patient history in reverse chronological order	TC-CLN-04
HMS-F-034	Patient views own records and bills	TC-CLN-05
HMS-F-040	Generate itemised bill	TC-BIL-01
HMS-F-041	Record payments, update bill status	TC-BIL-02
HMS-F-042	Generate printable receipt	TC-BIL-03
HMS-F-050	Daily appointment and revenue reports	TC-RPT-01
HMS-F-060	Create, deactivate and assign roles to users	TC-ADM-01
HMS-F-061	View audit log with filters	TC-ADM-02
HMS-NF-001	Patient search within 2 seconds for 95% of requests	TC-PERF-01
HMS-NF-002	Appointment booking within 3 seconds for 95% of requests	TC-PERF-02
HMS-NF-003	200 concurrent sessions for 30 minutes	TC-PERF-03
HMS-NF-004	99.5% availability, 06:00 to 22:00	TC-AVL-01
HMS-NF-005	Receptionist registers a patient in 3 minutes or less	TC-USA-01
HMS-NF-006	Automated encrypted backup every 24 hours	TC-REL-01
HMS-NF-007	Client runs on Windows 10/11 and Ubuntu 22.04	TC-PORT-01
HMS-NF-008	At least 70% unit-test line coverage per service module	TC-MNT-01
HMS-NF-009	90% of core tasks completed without help (5 users)	TC-USA-01
HMS-SEC-001	Salted password hashes, password strength rules	TC-SEC-01
HMS-SEC-002	TLS 1.2+ for all communication	TC-SEC-02
HMS-SEC-003	10-minute session timeout, token invalid after logout	TC-SEC-03
HMS-SEC-004	Server-side validation, parameterised queries	TC-SEC-04
HMS-SEC-005	Audit record for every login, view, creation and update	TC-SEC-05
HMS-SEC-006	AES-256 encryption at rest for patient fields	TC-SEC-06
HMS-SEC-007	Server-side authorisation on every API call	TC-SEC-07, TC-RBAC-01
HMS-SEC-008	No passwords or clinical content in audit records; Administrator-only read	TC-SEC-05

Coverage: 25 functional + 9 non-functional + 8 security requirements = 42 requirements, covered by 40 test cases.

14. Test Metrics & Reporting

- % of test cases executed, % passed / failed / blocked
- Defect density (defects per module) and defect aging (days open)
- Requirement coverage (requirements with a passed test / total requirements)
- Unit-test line coverage per module (target 70% or more)
- Performance: 95th percentile response time and error rate per load test

Reports: daily execution status during the execution window, and a final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
QA Lead	Daksha S G (PES1UG24CS137)	
Developer	Dhruv Swatantramath (PES1UG24CS155)	
Product Owner	Navya Suresh (PES1UG24CS904)	
Course Instructor	Dr. Ashok Kumar Patil	

Appendix A: Test Cases

The table lists 40 test cases: 25 functional cases, 8 non-functional cases and 7 security cases (the minimum requested is 7-10). Each case has a stable ID used in the traceability matrix. Dates are relative to the day of execution; user names and data are from the dummy test data set.

Test case ID	Requirement	Type	Priority	Preconditions	Steps and test data	Expected result
TC-AUTH-01	HMS-F-001	Functional	High	User dr.rao exists and is active; server running.	Open login screen. Enter username "dr.rao" and the correct password.	Login succeeds and the Doctor dashboard is shown.
TC-AUTH-02	HMS-F-004	Functional / Negative	High	User dr.rao exists; no user named nouser.	Enter username "dr.rao" with password "wrong123"; repeat with unknown username "nouser".	Both attempts show the same message "Invalid username or password"; no session is created.
TC-AUTH-03	HMS-F-002	Functional / Negative	High	dr.rao is active, not locked, failed-attempt counter at 0.	Enter a wrong password 5 times for "dr.rao", then enter the correct password immediately.	Account is locked; message shows lock for 15 minutes; correct password is refused until lock expires.
TC-RBAC-01	HMS-F-003, HMS-SEC-007	Security / Functional	High	Receptionist account with a valid token; SRS role-permission matrix available.	(a) Log in as Receptionist and call the revenue report API (GET /api/v1/reports/revenue) directly with the valid token. (b) Using the SRS role-permission matrix, call at least 5 other functions that the matrix marks as not permitted for the logged-in role (one per role).	(a), (b) HTTP 403 "Access denied"; no data returned; each event written to the audit log.
TC-PAT-01	HMS-F-010	Functional	High	Logged in as Receptionist; patient Asha Nair not yet registered.	Register patient: Asha Nair, DOB 14-03-1990, Female, phone 9876543210, address "12 MG Road, Bengaluru".	Record saved; unique ID in format PID-nnnnnnnn displayed; patient appears in search.
TC-PAT-02	HMS-F-011	Functional / Boundary	High	Logged in as Receptionist; registration form open.	Submit registration with (a) empty phone, (b) 9-digit phone, (c) 11-digit phone, (d) DOB 01-01-2030.	Each submission rejected with a message naming the field; nothing saved.
TC-PAT-03	HMS-F-012	Functional	Medium	Logged in as Receptionist; patient from TC-PAT-01 saved.	Register a second patient with the same name, DOB and phone as TC-PAT-01.	Duplicate warning is shown; record is saved only if the user confirms.
TC-PAT-04	HMS-F-013	Functional	High	Logged in as Receptionist; Asha Nair registered; PID-00000000 does not exist.	Search by (a) exact ID, (b) partial name "Ash", (c) phone, (d) ID "PID-00000000" that does not exist.	(a) exactly 1 result; (b), (c) all matches listed; (d) "No patient found".
TC-PAT-05	HMS-F-014	Functional	Medium	Logged in as Receptionist; Asha Nair registered.	As Receptionist change phone of Asha Nair to 9123456780 and save.	New phone shown; history shows old value, timestamp and user.
TC-APT-01	HMS-F-020	Functional	High	Logged in as Receptionist; Dr. Rao's slot free tomorrow; Asha Nair registered.	Book Dr. Rao, tomorrow 10:00-10:30 for Asha Nair.	Appointment ID returned, status "Confirmed", slot no longer shown as free.

Test case ID	Requirement	Type	Priority	Preconditions	Steps and test data	Expected result
TC-APT-02	HMS-F-021	Functional / Negative	High	TC-APT-01 done; a second patient registered.	Book the same doctor and slot as TC-APT-01 for another patient.	Booking refused; message states slot taken and lists next 3 available slots.
TC-APT-03	HMS-F-022	Boundary	Medium	Logged in as Receptionist; Dr. Rao's schedule defined for the dates used.	Try booking for (a) yesterday, (b) today 1 hour earlier than now, (c) day 60, (d) day 61 from today.	(a), (b), (d) rejected with reason; (c) accepted.
TC-APT-04	HMS-F-023	Boundary	Medium	Two confirmed appointments exist, starting in 3 h and in 1 h 59 min.	Cancel an appointment that starts in (a) 3 hours, (b) 1 hour 59 minutes.	(a) cancelled and slot freed; (b) rejected with "Cannot change within 2 hours".
TC-APT-05	HMS-F-024	Functional	Medium	Logged in as Dr. Rao; appointments booked out of time order, plus one for another doctor.	Doctor opens "My schedule" for a day with appointments booked out of time order.	Appointments listed sorted by start time; only own patients shown.
TC-CLN-01	HMS-F-030	Functional	High	Logged in as Doctor; confirmed appointment exists.	Doctor opens a confirmed appointment and saves symptoms "fever", diagnosis "viral infection", notes.	Visit saved with doctor ID and timestamp, linked to the appointment.
TC-CLN-02	HMS-F-031	Functional	High	Visit from TC-CLN-01 saved.	Add prescription: Paracetamol 500 mg, 3 times a day, 5 days.	Item saved on the visit and shown in the prescription list.
TC-CLN-03	HMS-F-032	Functional / Negative	High	Patient has allergy "Penicillin" recorded; Doctor logged in with an open visit.	Patient has recorded allergy "Penicillin". Doctor prescribes Amoxicillin.	Allergy warning displayed; item saved only after an override reason is entered.
TC-CLN-04	HMS-F-033	Functional	High	Logged in as Doctor; patient with 3 visits on different dates.	Open history of a patient with 3 visits on different dates.	Visits listed newest first with diagnoses and prescriptions.
TC-CLN-05	HMS-F-034	Functional / Negative	Medium	Logged in as Patient Asha Nair with at least 2 visits and 1 bill; another patient has records.	Log in as Patient Asha Nair; open own history and bills. Then call GET /api/v1/patients/{another patient ID}/history with the same token.	Only own visits, prescriptions and bills are shown, newest first; request for another patient returns HTTP 403 "Access denied".
TC-BIL-01	HMS-F-040	Functional	High	Logged in as Billing Officer; completed visit exists; tax rate set to 5%.	Create bill: consultation 500 + procedure 300, tax rate 5%.	Subtotal 800.00, tax 40.00, total 840.00.
TC-BIL-02	HMS-F-041	Functional / Boundary	High	Unpaid bill of 840.00 from TC-BIL-01.	On the 840.00 bill pay (a) 400 by UPI, (b) 500, (c) 440.	(a) status "Partially Paid", balance 440; (b) rejected as over-payment; (c) status "Paid", balance 0.
TC-BIL-03	HMS-F-042	Functional	Medium	Payment (a) of TC-BIL-02 recorded.	Generate receipt after payment (a) of TC-BIL-02.	Printable receipt with unique receipt number, amount, mode and date.

Test case ID	Requirement	Type	Priority	Preconditions	Steps and test data	Expected result
TC-RPT-01	HMS-F-050	Functional	Medium	Logged in as Administrator; payments recorded in the last 7 days.	As Administrator generate the revenue report for the last 7 days and export as CSV and PDF.	Totals equal the sum of recorded payments; both files download and open.
TC-ADM-01	HMS-F-060	Functional	High	Logged in as Administrator.	Administrator creates user "nurse1" with role Receptionist, then deactivates it; user then tries to log in.	Account created with role; after deactivation login is refused.
TC-ADM-02	HMS-F-061	Functional	Medium	Logged in as Administrator; audit entries for dr.rao exist today.	Administrator filters audit log by user "dr.rao" and today's date.	Only matching entries are shown with user, action, record ID and timestamp.
TC-PERF-01	HMS-NF-001	Performance	Medium	Test database holds 100,000 patients; JMeter script ready.	JMeter: 100 virtual users run patient searches for 10 minutes on a database of 100,000 patients.	95th percentile response time is 2 s or less.
TC-PERF-02	HMS-NF-002	Performance	Medium	20 doctors with free slots; JMeter script ready.	JMeter: 100 virtual users book appointments on distinct free slots for 10 minutes.	95th percentile response time is 3 s or less; no duplicate bookings.
TC-PERF-03	HMS-NF-003	Load	Medium	Test database and mixed-workload JMeter script ready.	JMeter: ramp to 200 concurrent sessions with a mixed workload for 30 minutes.	Error rate is below 1%; no crash.
TC-AVL-01	HMS-NF-004	Reliability	High	Staging system deployed; minute health-check configured.	Monitor the staging system for 7 days between 06:00 and 22:00 with a health-check every minute.	Measured availability is 99.5% or more.
TC-USA-01	HMS-NF-005, HMS-NF-009	Usability	Medium	5 trained receptionists; test data and task sheet ready.	5 trained receptionists each register a patient and book an appointment without help; time each run.	Registration takes 3 minutes or less; at least 90% of tasks completed unaided.
TC-REL-01	HMS-NF-006	Recovery	High	Backup job configured; test database populated.	Trigger the nightly backup, delete the test database and restore from the backup; time the restore.	Data restored with at most 24 h loss; restore completes within 4 h.
TC-PORT-01	HMS-NF-007	Compatibility	Low	Windows 11 (MSVC 2019) and Ubuntu 22.04 (GCC 11) build machines available.	Build and run the client on Windows 11 (MSVC 2019) and Ubuntu 22.04 (GCC 11); execute the smoke suite.	Both builds succeed and all smoke tests pass.
TC-MNT-01	HMS-NF-008	Maintainability	Medium	Unit tests exist for all service modules; gcov enabled.	Run the unit-test suite with gcov/lcov for every service module.	Line coverage is 70% or more per module.
TC-SEC-01	HMS-SEC-001	Security	High	Administrator logged in; read access to the credentials table.	(a) Create a user with password "abc"; (b) inspect the credentials table.	(a) rejected by the password policy; (b) only salted hashes, no plaintext.
TC-SEC-02	HMS-SEC-002	Security	High	Server running with TLS configured; testssl.sh installed.	Connect over plain HTTP; scan with a TLS scanner (e.g. testssl.sh).	HTTP refused; only TLS 1.2 or higher is accepted.
TC-SEC-03	HMS-SEC-003	Security	High	Valid session for any user.	Stay idle for 10 minutes then call an API; log out and replay the old token.	Both calls return 401 and require a new login.

Test case ID	Requirement	Type	Priority	Preconditions	Steps and test data	Expected result
TC-SEC-04	HMS-SEC-004	Security	High	Test database with dummy data; ZAP and sqlmap available.	Enter ' OR '1'='1 in login and search fields; enter <script>alert(1)</script> in a notes field.	Input rejected or stored as inert text; no data leaked; no script runs.
TC-SEC-05	HMS-SEC-005, HMS-SEC-008	Security	High	Audit logging enabled; a non-admin account available.	Log in, view a record and update it; then inspect the audit log and the log files.	Each action has an audit entry; no passwords or clinical content in logs; non-admin cannot read them.
TC-SEC-06	HMS-SEC-006	Security	High	SQL client access to the test database.	Query the database directly for patient name and diagnosis columns.	Values are ciphertext, not readable text.
TC-SEC-07	HMS-SEC-007	Security	High	Logged in as Patient; a second patient exists.	As Patient, call GET /api/v1/patients/{another patient ID} with a valid token.	HTTP 403; no data returned.